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Education Project

Socioeconomic Factors and High School ACT/SAT Scores

Abstract

This project asks how well socioeconomic factors predict a high school's average ACT score and whether charter status adds information beyond them. I combined 2016 EdGap.org data (census-tract income, unemployment, adult educational attainment, family structure, and the share of students eligible for free or reduced-price lunch) with school records from the National Center for Education Statistics (NCES). After cleaning, the sample contained 7,227 high schools in 20 states. OLS regression explained 62.8% of score variation; the reduced model's mean five-fold cross-validated R^2 was 0.600. Lunch eligibility alone explained 61.4%. Charter schools scored 1.34 points lower before adjustment, but the gap declined to −0.19 after socioeconomic adjustment (95% CI −0.42 to 0.05) and −0.10 after adding state. These school-level associations do not establish causation.

1. Introduction

Differences in school average ACT/SAT scores may reflect unequal educational opportunity rather than school quality alone. This project asks how well five socioeconomic measures predict average scores, which predictor is most informative, and whether charter and non-charter high schools differ after socioeconomic and state adjustment. I hypothesized that SES would explain most of the between-school variation and that most of any raw charter gap would reflect the communities that charter schools serve.

Two public data sets were combined. The EdGap.org data (EdGap.org, 2016) cover 7,986 schools and contain the school's average ACT score (SAT averages are converted to ACT equivalents), the share of students eligible for free or reduced-price lunch, and four census-tract variables from the American Community Survey (U.S. Census Bureau, n.d.): median household income, unemployment rate, share of adults with a college degree, and share of children in married-couple families. Scores come from state education departments or other public releases. The NCES Common Core of Data school file for 2016-17 has 102,183 records and 65 variables, of which we used the school ID, state, school level, and charter status. The files were joined on the NCES school ID, restricted to high schools, cleaned, imputed for a few missing values, leaving 7,227 schools.

2. Theoretical background

- **Linear regression**

OLS estimates $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \varepsilon$ by minimizing the sum of squared residuals. A coefficient is the expected change in y per one-unit change in its predictor, holding the others fixed. Fit is summarized by R^2 (the share of variance explained), adjusted R^2, and the mean absolute error (MAE) and root mean squared error (RMSE), which are in the units of y. Standardized predictors allow coefficient comparisons, while correlated predictors can increase uncertainty.

- **Comparing models**

For nested models, an F-test asks whether the q extra terms reduce the residual sum of squares by more than chance: $F = [(RSS_r - RSS_f)/q] / [RSS_f/(n - p)]$, where q is the number of additional

terms and p includes the full model's intercept. Five-fold cross-validation assesses predictions on schools withheld from fitting.

- **Missing data and adjustment**

Iterative imputation predicts incomplete variables from other predictors (van Buuren & Groothuis-Oudshoorn, 2011; Pedregosa et al., 2011). Charter indicators estimate adjusted group differences, and state indicators account for mean differences across states. School-level associations need not hold for individuals (Robinson, 1950).

3. Methodology

All steps are in code/Education.ipynb (github.com/giangtranq/education) and use Python with pandas, NumPy, statsmodels and scikit-learn.

- **Data preparation**

- Read EdGap_data.xlsx (school ID as text) and the NCES CSV (unicode_escape encoding). Keep the NCES columns for year, school ID, state, zip code, school type, level, and charter status, and rename all columns to snake_case.
- Left-join the NCES records onto EdGap on the school ID (NCESSCH) after converting both IDs to the same type. 7,898 of the 7,986 EdGap schools matched.
- Keep schools with level "High" (7,230). No duplicate rows were found.
- Set ACT averages below 1 (3 schools; the valid range is 1-36) and negative lunch shares (20 schools) to missing.
- Drop the 3 schools with no ACT average, leaving 7,227. Impute the five numerical predictors jointly with IterativeImputer and its default BayesianRidge estimator. Each predictor has at most 20 missing values after cleaning.
- Derive is_charter (1 = charter, 0 = non-charter, missing when charter status is "Not applicable", 199 schools) and export data/education_clean.csv.

- **Exploratory analysis**

I examined summary statistics, Pearson correlations, pair plots, and box plots (outliers by the 1.5 × IQR rule). Outliers were retained because they appeared to be real values.

- **Models of ACT scores**

All models are OLS fits (statsmodels formula API) of the average ACT score; shares are on a 0-1 scale and income is in dollars. We fitted a model for each single predictor, lunch share plus its square, lunch share plus median income, all five predictors (full), and unemployment rate, college degree share, and lunch share (reduced). R^2, MAE, and RMSE were computed on all 7,227 schools. Nested models were compared with F-tests (anova_lm): linear vs. quadratic lunch share and reduced vs. full. The reduced model was also evaluated with 5-fold cross-validation (LinearRegression, default unshuffled folds, scoring R^2). To compare effect sizes, the reduced predictors were standardized (StandardScaler) and the model refitted. Residuals were plotted against fitted values.

- **Charter comparison**

Using the 7,028 schools with charter status "Yes" or "No", we compared group means and fitted (C1) ACT ~ is_charter, (C2) C1 plus the five SES variables, and (C3) C2 plus state indicators. We

also tested whether adding is_charter to the reduced model improves it (F-test). Tests are two-sided with $\alpha = 0.05$.

4. Computational results

	mean	std	min	max	correlation_with_act
average_act	20.299	2.508	12.363	32.363	1.000
rate_unemployment	0.096	0.057	0.000	0.590	-0.433
percent_college	0.573	0.165	0.091	1.000	0.458
percent_married	0.640	0.191	0.000	1.000	0.440
median_income	52745.418	24359.994	1091.668	226181.000	0.460
percent_lunch	0.411	0.234	0.000	0.999	-0.784

Table 1: Descriptive statistics and correlation with the average ACT score (n = 7,227)

Schools per state ranged from 24 (Delaware) to 913 (Texas). The largest correlations between predictors were college degree share with median income (0.71) and median income with married-couple share (0.62).

	R-squared	MAE	RMSE
Lunch share	0.614	1.169	1.559
Lunch share + median income	0.619	1.161	1.548
Lunch share + lunch share squared	0.616	1.165	1.555
Five predictors (full)	0.628	1.145	1.530
Unemployment, college, lunch (reduced)	0.628	1.145	1.530

Table 2: Fit of models predicting the average ACT score (7,227 schools, MAE and RMSE in ACT points)

The single-predictor R^2 values were 0.188 (unemployment rate), 0.210 (college degree share), 0.193 (married-couple share), and 0.211 (median income). Mean cross-validated R^2 was 0.600 for the reduced model and 0.597 for the full model. Reduced versus full gave $F(2, 7221) = 0.215$, $p = 0.807$; adding the quadratic lunch term gave $F(1, 7224) = 34.59$, $p = 4.3 \times 10^{-9}$. The lunch-only slope was −8.390 (95% CI −8.544 to −8.237). Residual skewness was 0.58 and kurtosis 5.8.

	Coefficient	95% CI lower	95% CI upper	Standardized coefficient	p-value
Intercept	22.640	22.439	22.841	NaN	0.0
rate_unemployment	-2.168	-2.901	-1.436	-0.123	0.0
percent_college	1.714	1.465	1.963	0.283	0.0
percent_lunch	-7.586	-7.768	-7.405	-1.777	0.0

Table 3: Reduced model coefficients (outcome: average ACT score)

In the full model, the married-couple share ($p = 0.577$) and median income ($p = 0.920$) were not significant. The condition number was 1.34×10^6 for the full model and 25.9 for the reduced model.

	charter	No	Yes
average_act	mean	20.361	19.020
	std	2.496	3.143
rate_unemployment	mean	0.095	0.139
	std	0.055	0.089
percent_college	mean	0.575	0.551
	std	0.164	0.194
percent_married	mean	0.645	0.492
	std	0.189	0.235
median_income	mean	53356.946	41810.166
	std	24470.042	21352.457
percent_lunch	mean	0.406	0.541
	std	0.234	0.282

Table 4: Charter and non-charter high schools, mean (SD)

	Charter coefficient	95% CI lower	95% CI upper	p-value	R-squared
No controls	-1.341	-1.724	-0.958	0.000	0.007
Socioeconomic controls	-0.186	-0.423	0.051	0.124	0.628
Socioeconomic controls + state	-0.095	-0.318	0.127	0.401	0.678
Reduced model + is_charter	-0.179	-0.415	0.058	0.138	0.628

Table 5: Estimated charter minus non-charter difference in the average ACT score (n = 7,028)

Of the 170 charter schools, 85 were in Florida (31), Michigan (27), and Texas (27). None were in North Carolina, Washington, or Wyoming.

5. Discussion

- **SES predicts school average ACT scores well**

The five SES variables explained 62.8% of the variance, and the cross-validated R^2 of 0.60 shows this is not an artifact of overfitting. A typical error is 1.1 points (MAE) to 1.5 points (RMSE), against a between-school standard deviation of 2.5 points. The remaining 37% of the variance probably reflects unmeasured factors such as school quality, student composition, test-taking rates.

- **Lunch eligibility dominates**

Lunch eligibility alone explained 61.4% of variation. A 10-percentage-point increase was associated with a 0.84-point lower ACT average in the single-predictor model. Its standardized reduced-model coefficient was six times the magnitude of college-degree share and 14 times unemployment. Income raised lunch-only R^2 by just 0.005. Overlapping socioeconomic information limits additional predictors' contributions; unlike tract income, lunch eligibility measures the enrolled student population.

- **Model form**

The quadratic lunch term was statistically significant but raised R^2 by only 0.002 and lowered MAE by 0.004 points. With 7,227 schools, trivial effects can be significant, so we kept the linear form. The reduced model fit as well as the full model ($p = 0.807$). Residual spread was constant,

but the right skew and heavy tails mean some schools score far above their predictions, so intervals that assume normal errors are approximate.

- **Charter schools**

Charter schools scored 1.34 points lower, about half a standard deviation, and serve more disadvantaged communities (54% of students eligible for subsidized lunch vs. 41%; median income about $11,500 lower). Adjusting for SES reduced the gap by 86% to -0.19 points ($p = 0.124$), and adding state reduced it to -0.10 points ($p = 0.401$). Adding is_charter to the reduced model did not significantly improve it ($p = 0.138$). These results support our hypothesis that most of the raw gap reflects the communities charter schools serve, but do not establish equivalence or causally explain the gap. The adjusted interval permits a disadvantage of approximately 0.42 points. Concentration in three states and student selection limit interpretation.

- **Limitations**

These are associations, not causal effects. The 20 states reflect EdGap's data availability, the data cover one year, and census tracts only approximate attendance areas. School averages cannot describe individual students, SAT-to-ACT conversion and varying participation add noise, and the hand-assembled EdGap data may contain errors. Imputation and the choice of the reduced model used all of the data, so the cross-validated R^2 may be slightly optimistic, although the reduced and full models scored almost the same (0.600 vs. 0.597).

6. Conclusion

Across 7,227 schools in 20 states, socioeconomic variables accounted for roughly 63% of the differences between schools, and the share of students eligible for subsidized lunch was by far the most informative single measure. The lower average scores of charter schools were largely accounted for by the more disadvantaged communities they serve, although the sample cannot exclude a small remaining difference. School averages should therefore be interpreted alongside socioeconomic context rather than as independent measures of school quality. Explaining the remaining 37% of the variation will require student-level data, measures of school resources, more states, and designs that can separate student selection from school effects.

7. References

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